I. Supporting Text

1. Experimental Procedures

1.1. Density Functional Theory

All possible bonding structures involving lithium (Li) ions interacting with various solvent molecules and anions were constructed and their ^{7}Li chemical shifts were specially focused on. Solvent molecules such as 1,2-dimethoxyethane (DME), dimethyl carbonate (DMC), 1,3-dioxolane (DOL), and ethylene carbonate (EC), along with anions including bis(fluorosulfonyl)imide anion (FSI^{-}), bis(fluorosulfonyl)imide ($TFSI^{-}$), PF_6^{-}, and NO_3^{-}, were considered. These species can interact with Li atoms through one or two O or F atoms, leading to the development of possible bonding structures (**Figure S1**). These structures were optimized by Gaussian 16 using Lee–Yang–Parr correlation functional (B3LYP) at a basis set of 6–311++G(d,p). Solvent effects were incorporated using the solvation model based on solute electron density (SMD), with specific solvent parameters defined as follows: DME (dielectric constant = 7.20, solvent radius = 4.19 Å), DMC (dielectric constant = 3.11, solvent radius = 3.79 Å), DOL (dielectric constant = 7.10, solvent radius = 3.71 Å), and EC (dielectric constant = 89.78, solvent radius = 3.81 Å).[1–3] The nuclear magnetic resonance (NMR) shielding tensors for ^{7}Li in the optimized structures were computed at the same theoretical level. The NMR shielding tensor of Li in the Li^{+}–$(H_2O)_3$ complex was selected as the reference, which is a choice frequently used in experimental NMR characterizations. Consequently, the ^{7}Li chemical shift for each bonding structure was calculated by subtracting the shielding tensor of Li in the specific structure from that of

the Li^+–$(H_2O)_3$ reference.

The bond energy between Li and its coordinating species was determined through a systematic computational approach. Initially, the Li bonding structures were optimized at the aforementioned B3LYP/6–311++G(d,p) level with the SMD solvent model, and their total energies were calculated. Subsequently, the Li nucleus was removed from each optimized structure, and the energy of the remaining solvent and anion components was computed. The bond energy between the Li nucleus and its bonded species was defined by the following expression:

$$\text{Bond energy} = E_{\text{Li}^+-\text{solvent}_x-\text{anion}_y} - E_{\text{Li}^+} - E_{\text{solvent}_x-\text{anion}_y} \quad (1)$$

where $E_{\text{Li}^+-\text{solvent}_x-\text{anion}_y}$ represents the total energy of the Li^+–solvent–anion bonding structure, E_{Li^+} the energy of the Li^+ nucleus, and $E_{\text{solvent}_x-\text{anion}_y}$ the energy of the solvent–anion structure after Li^+ removal, respectively. The x and y range from 0 to 4, corresponding to the number of solvent molecules and anions, respectively.

To characterize the structural properties of the Li bonding structures, the lengths of all Li–O and Li–F bonds within the optimized structures were measured. These distances were then averaged to determine the mean bond length for each structure.

The electron localization function (ELF) surrounding Li in the optimized bonding structures was analyzed using Multiwfn (version 3.8dev).[4] ELF serves as a valuable tool for visualizing the degree of electron localization in three-dimensional space, offering insights into bonding characteristics. Following the definition proposed by Becke and Edgecombe in 1990, ELF is expressed as:[5]

$$ELF(\boldsymbol{r}) = \frac{1}{1+\chi_\sigma^2(\boldsymbol{r})} \quad (2)$$

where

$$\chi_\sigma^2(\boldsymbol{r}) = \frac{D_\sigma(r)}{D_\sigma^0(r)} \quad (3)$$

represents the ratio of local electron localization to that of a uniform electron gas, mapping the localization to a dimensionless index within the range $0 \leqslant \mathrm{ELF} \leqslant 1$. The term $D_\sigma(\boldsymbol{r})$ is defined as:

$$D_\sigma(\boldsymbol{r}) = \tau_\sigma(\boldsymbol{r}) - \frac{1}{4}\frac{\left(\nabla\rho_\sigma(\boldsymbol{r})\right)^2}{\rho_\sigma(\boldsymbol{r})} \quad (4)$$

where $\rho_\sigma(\boldsymbol{r})$ and $\tau_\sigma(\boldsymbol{r})$ is the electron spin density and the electron kinetic energy density, respectively. For a uniform electron gas with spin density $\rho_\sigma(\boldsymbol{r})$, the reference term simplifies to:

$$D_\sigma^0(r) = \frac{3}{5}(6\pi^2)^{\frac{2}{3}}\rho_\sigma^{\frac{5}{3}}(\boldsymbol{r}) \quad (5)$$

The ELF deformation factor, denoted as ϕ, was introduced to quantify deviations from spherical symmetry in the electron localization around Li, defined as:

$$\phi = \frac{A_{\mathrm{Li}}}{A_{Sphere}} \quad (6)$$

where A_{Li} is the surface area of the ELF isosurface at a value of 0.75, and A_{Sphere} the surface area of a sphere possessing the same volume as the ELF isosurface.

1.2. Molecular Dynamics Simulations

Molecular dynamics (MD) simulations were conducted using the large scale atomic/molecular massively parallel simulator (LAMMPS) code (ver. 29 Sep 2021).[6] 48 electrolyte models containing 200 lithium salt ($LiPF_6$, $LiNO_3$, LiFSI, and LiTFSI)

pairing with 1640, 760, and 280 organic solvent molecules (DME, DMC, DOL, and EC) were constructed, respectively, denoted as LC, MC, and HC electrolytes. The initial atomic coordinates were generated with PACKMOL (ver. 20.3.5), and the MD snapshots were visualized via VESTA (ver. 3.5.7).[7,8]

The MD simulation process consisted of several consecutive phases. Firstly, all electrolyte models were equilibrated in an isothermal-isobaric (constant NPT) ensemble using the Parrinello–Rahman barostat for 0.5 ns to maintain a temperature of 298 K and a pressure of 1 atm. After that, the electrolytes were heated from 298 to 420 K within 0.5 ns, maintained at 450 K for 0.5 ns, and then cooled from 450 to 298 K within 0.5 ns, followed by equilibration at 298 K for another 2 ns. A 5 ns production run was finally conducted in the canonical (constant NVT) ensemble under the Nose–Hoover thermostat. The final 5 ns were sampled to identify and analyze the Li bonding structures.

All Li bonding structures from the final 20 fs of each MD simulation were extracted and input into the machine learning model to predict ^{7}Li chemical shifts. The frequency of predicted shifts within the range of −4 to 3 ppm was calculated in 0.1 ppm intervals. Using the midpoints of these intervals and their corresponding frequencies, a cubic spline interpolation was applied to generate the NMR intensity distribution as a function of chemical shift for the LC, MC, and HC solvent/anion electrolytes.

1.3. Machine Learning Methods

To enable efficient prediction of NMR chemical shifts based on the Li coordination

environment, machine learning models were developed. The dataset consisted of 183 samples derived from the aforementioned DFT calculations, partitioned into training (80%) and test (20%) sets via stratified sampling.

The feature set comprised five variables: (1) the coordination number of anions (CNA), (2) the coordination number of solvent molecules (CNS), (3) the number of coordinating anions (NCA), (4) the number of coordinating solvent molecules (NCS), and (5) a weighted-average HOMO–LUMO energy gap (GAP) of the coordinating species. These features were selected based on their relevance to the Li ion's local environment, which can be readily extracted from MD trajectories, and the need to account for systematic biases arising from chemical differences among coordinating species by incorporating electronic structure-related properties.

According to NMR chemical shift theory, the shielding tensor is given by:[9]

$$\sigma = \mathrm{Av}_\lambda \sigma_\lambda = \frac{e^2}{3mc^2} < \psi_0 \left| \sum_k \frac{1}{r_k} \right| \psi_0 > -2\mathrm{Av}_\lambda \sum_n \frac{1}{E_n - E_0} \left(\begin{array}{l} < \psi_0 \left| \sum_k m_{zk}^0 \right| \psi_n >< \psi_n \left| \sum_k \frac{m_{zk}^0}{r_k^3} \right| \psi_0 > \\ + < \psi_0 \left| \sum_k m_{zk}^0 \right| \psi_n >< \psi_n \left| \sum_k \frac{m_{zk}^0}{r_k^3} \right| \psi_0 > \end{array} \right) \quad (7)$$

where

$$m_{zk}^0 = -\frac{e\hbar}{2mci} \left(x_k \frac{\partial}{\partial y_k} - y_k \frac{\partial}{\partial x_k} \right) \quad (8)$$

Av_λ denotes averaging over the λ direction, $E_n (n > 0)$ are the energies of excited states, E_0 the ground state energy, and $\psi_n (n > 0)$, and ψ_0 the wavefunctions of the excited and ground states, respectively. Variations in the energy differences between

ground and excited states, influenced by different solvents and anions, are dominated by the gap between the lowest excited state and the ground state $E_1 - E_0$. Given the established positive correlation between the system's HOMO–LUMO gap and $E_1 - E_0$, the descriptor GAP was defined as:

$$\mathrm{GAP} = \frac{\mathrm{NCA}}{\mathrm{NCA} + \mathrm{NCS}}(\mathrm{LUMO} - \mathrm{HOMO})_{\mathrm{anion}} + \frac{\mathrm{NCS}}{\mathrm{NCA} + \mathrm{NCS}}(\mathrm{LUMO} - \mathrm{HOMO})_{\mathrm{solvent}} (9)$$

where NCA and NCS are the coordination numbers of ions and solvent molecules, respectively. The weighted average approximates the system's HOMO–LUMO gap, with computational validation confirming a strong positive correlation ($R^2 \approx 0.75$, see **Figure S17**), justifying its inclusion as a feature.

To validate the selected feature set, the degree of multicollinearity among the five descriptors (CNA, CNS, NCA, NCS, and GAP) was assessed using a Pearson Correlation Coefficient (PCC) analysis (**Figure S29**). This statistical method was employed to quantify the linear relationship between each pair of features and ensure their relative independence. As a result, the off-diagonal PCC values are all well below the common threshold of 0.9, indicating that no strong inter-feature correlations exist. This lack of strong multicollinearity confirms the statistical independence of the descriptors, validating their suitability for the development of the machine learning models.

Two machine learning algorithms were employed: Support Vector Regression (SVR) and Gradient Boosting Regression (GBR), with their predictions combined using a voting ensemble approach. Hyperparameters were optimized through grid

search with 10-fold cross-validation in the training set. For GBR, the optimal parameters were: learning_rate = 0.1, max_depth = 3, n_estimators = 300, max_leaf_nodes = 6, yielding $R^2 \approx 0.91$ on the validation set. For SVR, the optimal parameters were: C = 10, epsilon = 0.1, kernel = 'rbf', achieving $R^2 \approx 0.92$ on the validation set. The ensemble model improved performance to $R^2 \approx 0.93$ on the test set.

Model performance was assessed using the root mean square error (RMSE) and coefficient of determination (R^2), defined as:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - y_i^{pred}\right)^2} \tag{10}$$

$$R^2 = \frac{\left(\sum_{i=1}^{n}(y_i - y_{mean}) \times \left(y_i^{pred} - y_{mean}^{pred}\right)\right)^2}{\sum_{i=1}^{n}(y_i - y_{mean})^2 \times \sum_{i=1}^{n}\left(y_i^{pred} - y_{mean}^{pred}\right)^2} \tag{11}$$

where y_i is the true value, y_i^{pred} the predicted value, and y_{mean} the mean of y_i, respectively. Additionally, Shapley Additive Explanations (SHAP) analyses were utilized to evaluate feature contributions, enhancing model interpretability.

To validate the effectiveness of the machine learning model developed from previous DFT data of Li-ion solvation structures, encompassing various anions, solvents, and Li^+, the ^7Li chemical shifts for all possible solvation structures in electrolytes were predicted. All Li bonding structures from the final 20 fs of each MD simulation were extracted and input into the machine learning model to predict ^7Li chemical shifts. The frequency of predicted shifts within the range of −4 to 3 ppm was

calculated in 0.1 ppm intervals. For each electrolyte environment, the predicted chemical shifts from these systems were averaged to obtain a representative value. Additionally, the frequency distribution of the predicted chemical shifts was analyzed across all simulations. Using the midpoints of these intervals and their corresponding frequencies, a cubic spline interpolation was applied to generate the NMR intensity distribution as a function of chemical shift for the LC, MC, and HC solvent/anion electrolytes. Furthermore, these predicted chemical shifts were averaged to determine the final predicted ^{7}Li chemical shift in the electrolyte.

II. Supplemental Figures

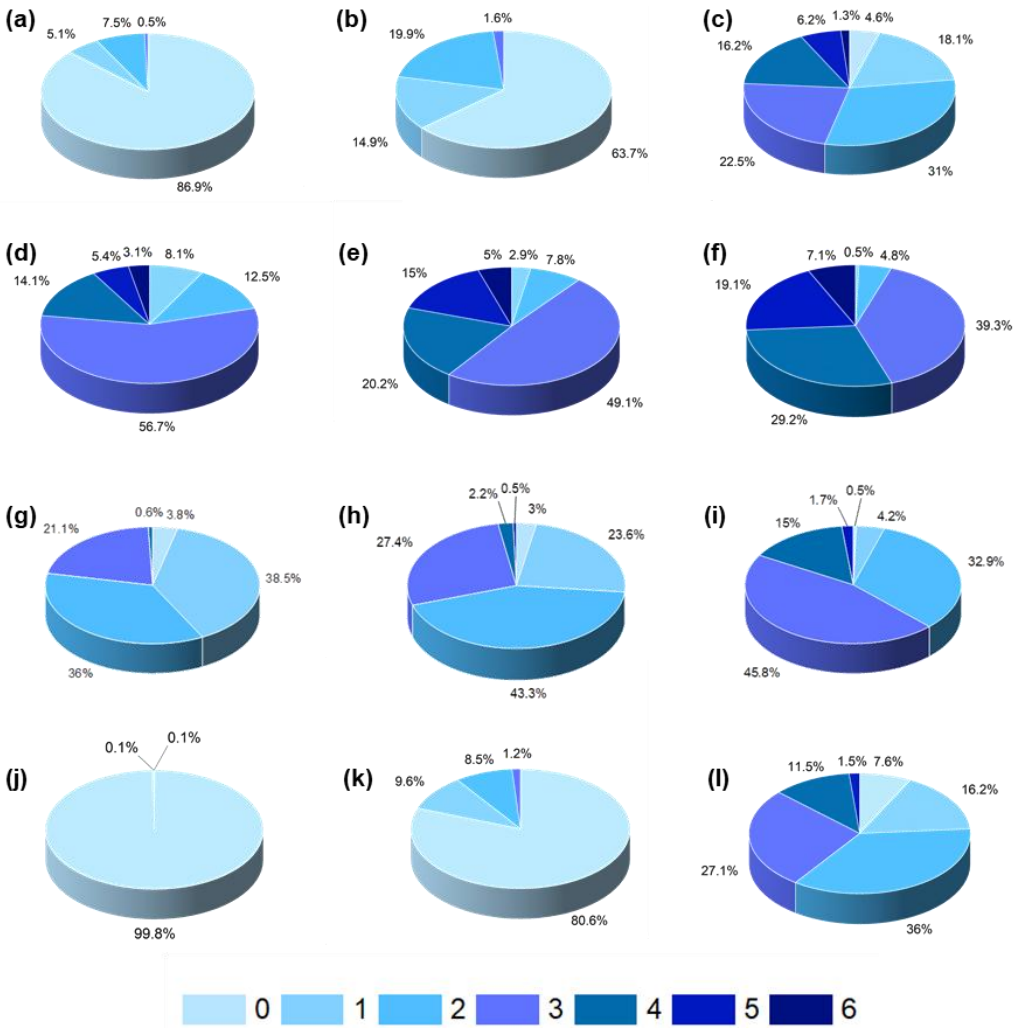


Figure S1. Bonding structure distributions of DME electrolytes with different salt concentrations. (a–c) Bonding structure distributions of (a) LC, (b) MC, and (c) HC DME/$LiPF_6$ electrolytes. (d–f) Bonding structure distributions of (d) LC, (e) MC, and (f) HC DME/$LiNO_3$ electrolytes. (g–i) Bonding structure distributions of (g) LC, (h) MC, and (i) HC DME/LiFSI electrolytes. (j–l) Bonding structure distributions of (j) LC, (k) MC, and (l) HC DME/LiTFSI electrolytes. The numbers 0 to 6 represent the number of anions coordinating with each Li^+.

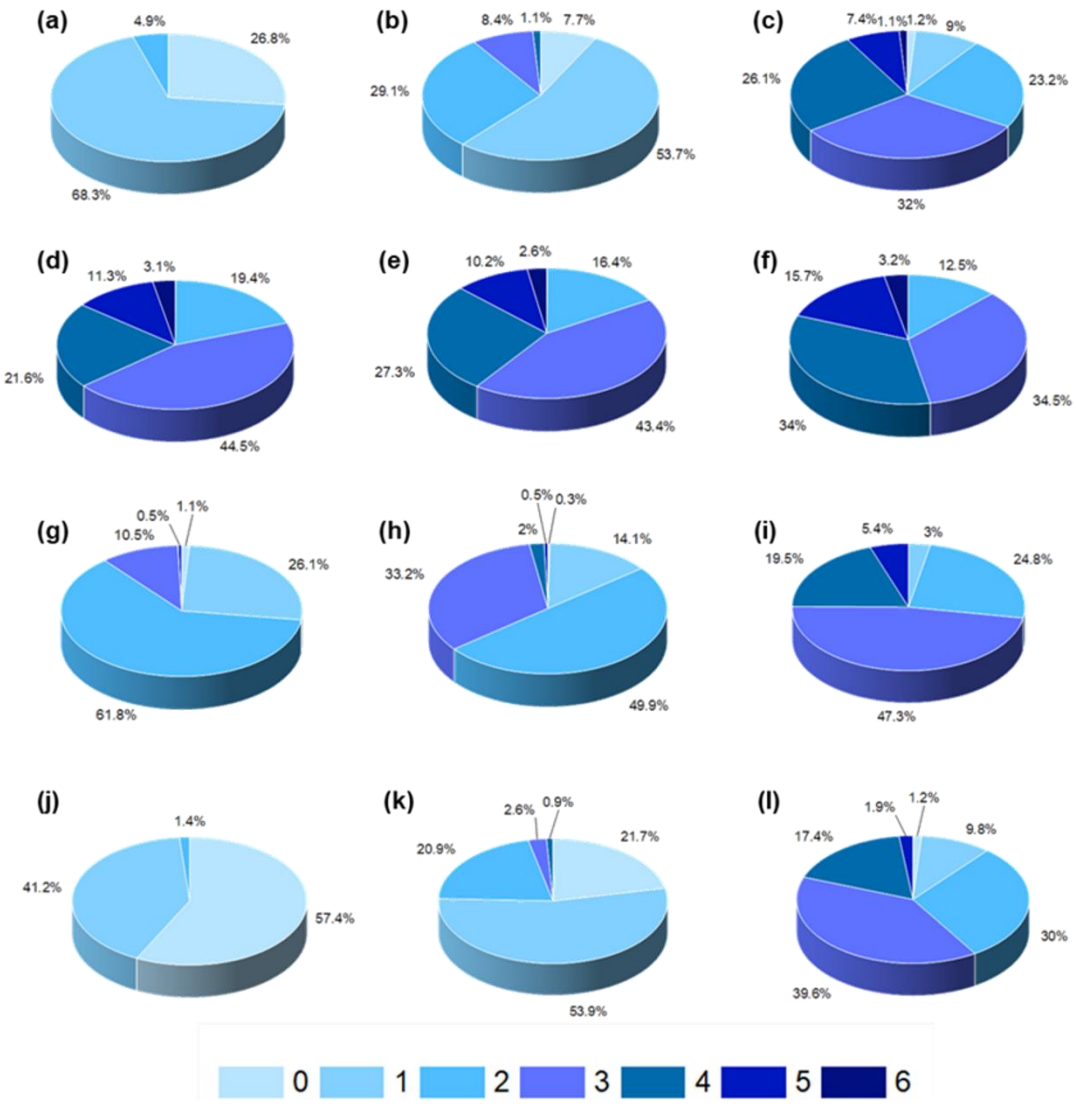


Figure S2. Bonding structure distributions of DMC electrolytes with different salt concentrations. (a–c) Bonding structure distributions of (a) LC, (b) MC, and (c) HC DMC/$LiPF_6$ electrolytes. (d–f) Bonding structure distributions of (d) LC, (e) MC, and (f) HC DMC/$LiNO_3$ electrolytes. (g–i) Bonding structure distributions of (g) LC, (h) MC, and (i) HC DMC/LiFSI electrolytes. (j–l) Bonding structure distributions of (j) LC, (k) MC, and (l) HC DMC/LiTFSI electrolytes. The numbers 0 to 6 represent the number of anions coordinating with each Li^+.

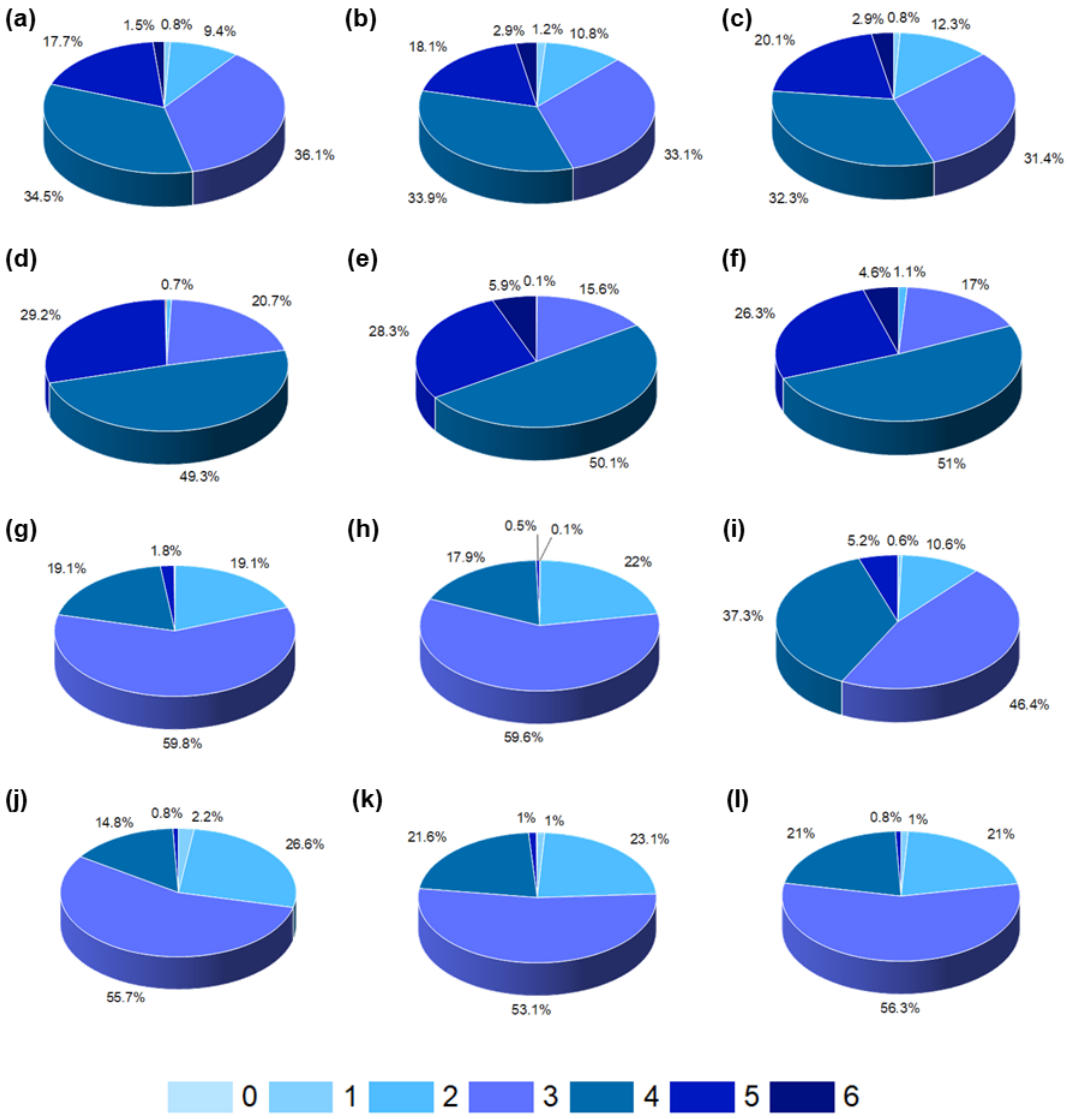


Figure S3. Bonding structure distributions of DOL electrolytes with different salt concentrations. (a–c) Bonding structure distributions of (a) LC, (b) MC, and (c) HC DOL/$LiPF_6$ electrolytes. (d–f) Bonding structure distributions of (d) LC, (e) MC, and (f) HC DOL/$LiNO_3$ electrolytes. (g–i) Bonding structure distributions of (g) LC, (h) MC, and (i) HC DOL/LiFSI electrolytes. (j–l) Bonding structure distributions of (j) LC, (k) MC, and (l) HC DOL/LiTFSI electrolytes. The numbers 0 to 6 represent the number of anions coordinating with each Li^+.

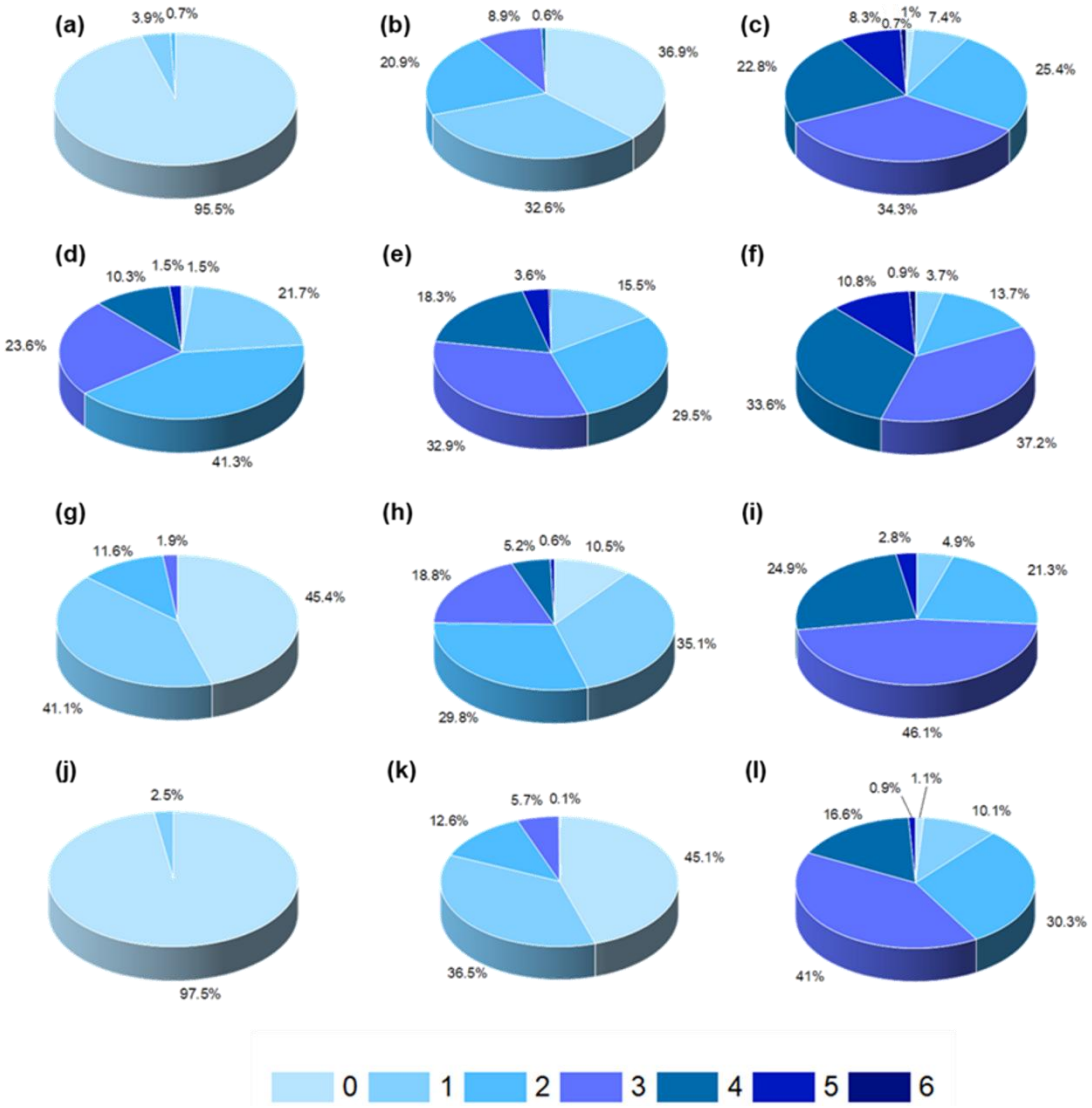


Figure S4. Bonding structure distributions of EC electrolytes with different salt concentrations. (a–c) Bonding structure distributions of (a) LC, (b) MC, and (c) HC EC/$LiPF_6$ electrolytes. (d–f) Bonding structure distributions of (d) LC, (e) MC, and (f) HC EC/$LiNO_3$ electrolytes. (g–i) Bonding structure distributions of (g) LC, (h) MC, and (i) HC EC/LiFSI electrolytes. (j–l) Bonding structure distributions of (j) LC, (k) MC, and (l) HC EC/LiTFSI electrolytes. The numbers 0 to 6 represent the number of anions coordinating with each Li^+.

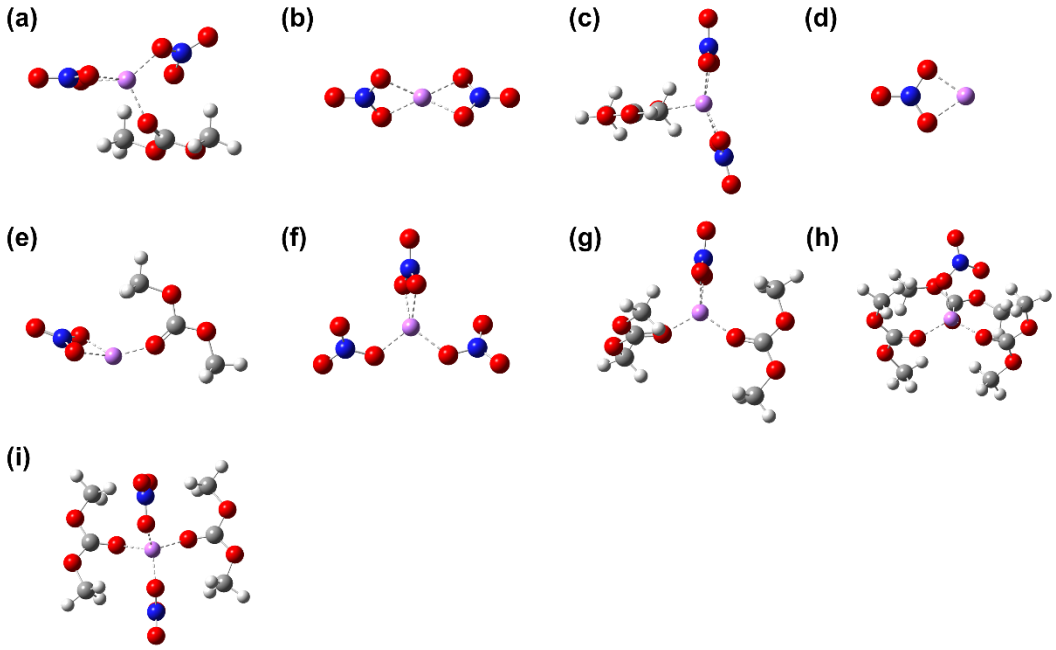


Figure S5. Optimized bonding structures of Li^+ and NO_3^- and/or DMC. (a) Li^+–$DMC_{1(1)}$–$NO_3^-{}_{2(2+1)}$, (b) Li^+–$NO_3^-{}_{2(2+2)}$, (c) Li^+–$DMC_{1(1)}$–$NO_3^-{}_{2(2+2)}$, (d) Li^+–$NO_3^-{}_{1(2)}$, (e) Li^+–$DMC_{1(1)}$–$NO_3^-{}_{1(2)}$, (f) Li^+–$NO_3^-{}_{3(2+1+1)}$, (g) Li^+–$DMC_{2(1+1)}$–$NO_3^-{}_{1(2)}$, (h) Li^+–$DMC_{3(1+1+1)}$–$NO_3^-{}_{1(1)}$, (i) Li^+–$DMC_{2(1+1)}$–$NO_3^-{}_{2(1+1)}$. The subscript of each label indicates the respective number of DMC and NO_3^-, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

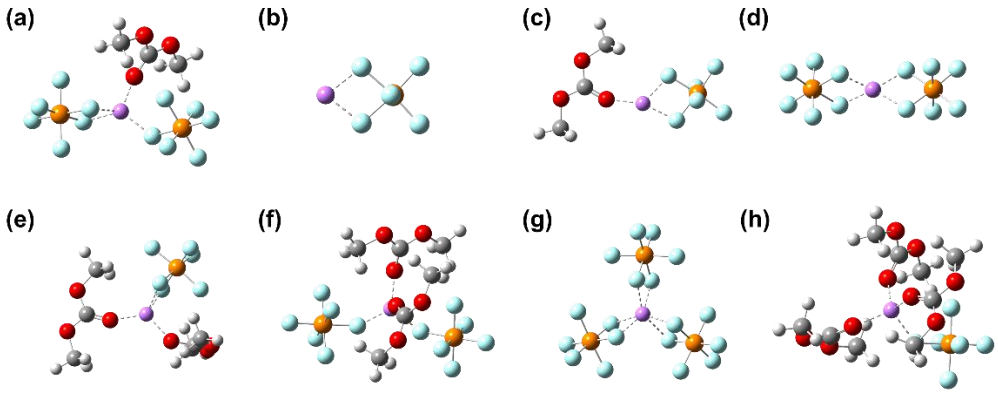


Figure S6. Optimized bonding structures of Li^+ and PF_6^- and/or DMC. (a) Li^+–$DMC_{1(1)}$–$PF_6^-{}_{2(2+1)}$, (b) Li^+–$PF_6^-{}_{1(2)}$, (c) Li^+–$DMC_{1(1)}$–$PF_6^-{}_{1(2)}$, (d) Li^+–$PF_6^-{}_{2(2+2)}$, (e) Li^+–$DMC_{2(1+1)}$–$PF_6^-{}_{1(2)}$, (f) Li^+–$DMC_{2(1+1)}$–$PF_6^-{}_{2(1+1)}$, (g) Li^+–$PF_6^-{}_{3(2+2+2)}$, (h) Li^+–$DMC_{3(1+1+1)}$–$PF_6^-{}_{1(1)}$. The subscript of each label indicates the respective number of DMC and PF_6^-, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

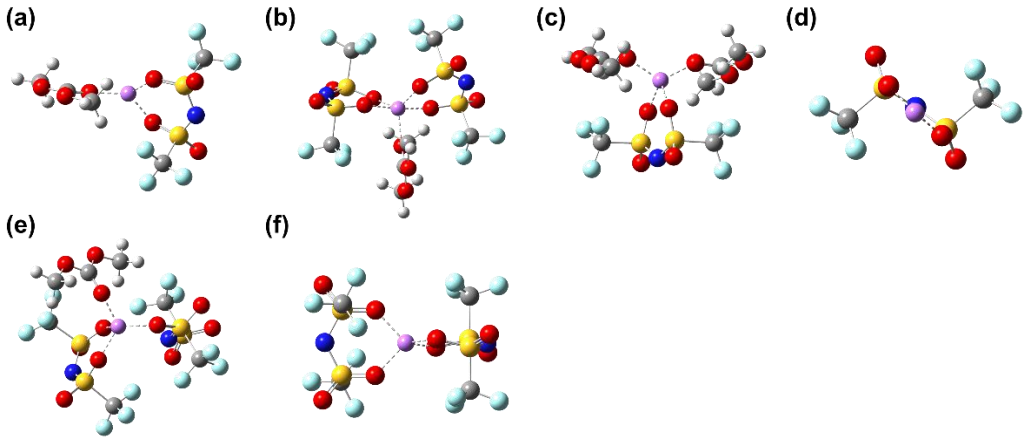


Figure S7. Optimized bonding structures of Li^+ and $TFSI^-$ and/or DMC. (a) Li^+–$DMC_{1(1)}$–$TFSI^-_{1(2)}$, (b) Li^+–$DMC_{1(1)}$–$TFSI^-_{2(2+2)}$, (c) Li^+–$DMC_{2(1+1)}$–$TFSI^-_{1(2)}$, (d) Li^+–$TFSI^-_{1(2)}$, (e) Li^+–$DMC_{1(1)}$–$TFSI^-_{2(2+1)}$, (f) Li^+–$TFSI^-_{2(2+2)}$. The subscript of each label indicates the respective number of DMC and $TFSI^-$, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

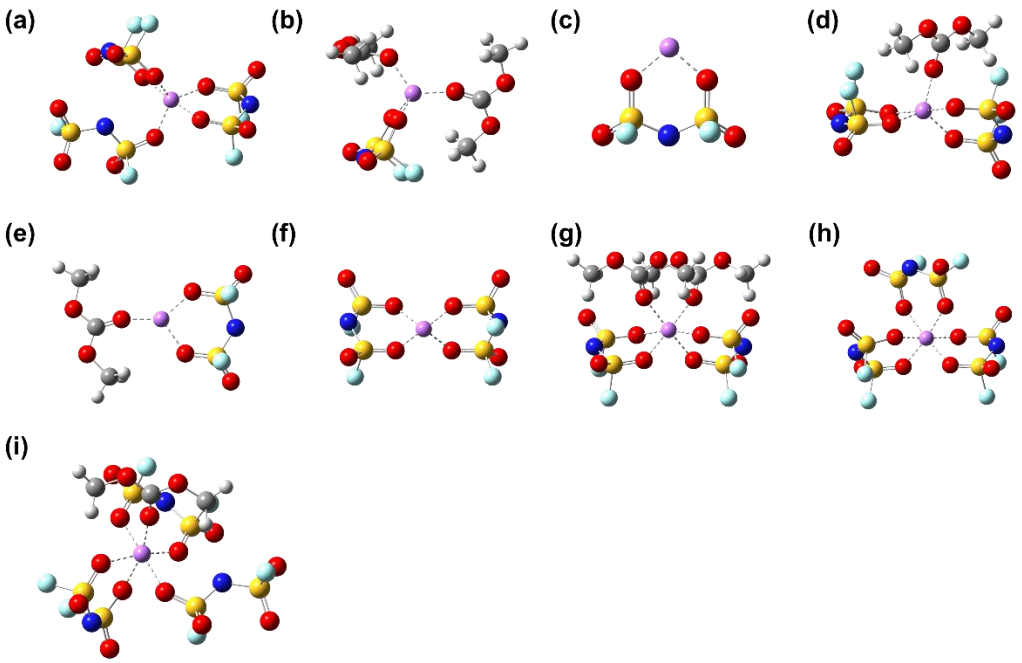


Figure S8. Optimized bonding structures of Li^+ and FSI^- and/or DMC. (a) Li^+–$FSI^-_{3(2+2+1)}$, (b) Li^+–$DMC_{2(1+1)}$–$FSI^-_{1(2)}$, (c) Li^+–$FSI^-_{1(2)}$, (d) Li^+–$DMC_{1(1)}$–$FSI^-_{2(2+2)}$, (e) Li^+–$DMC_{1(1)}$–$FSI^-_{1(2)}$, (f) Li^+–$FSI^-_{2(2+2)}$, (g) Li^+–$DMC_{2(1+1)}$–$FSI^-_{2(2+2)}$, (h) Li^+–$FSI^-_{2(2+2+2)}$, (i) Li^+–$DMC_{1(1)}$–$FSI^-_{3(2+2+1)}$. The subscript of each label indicates the respective number of DMC and FSI^-, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

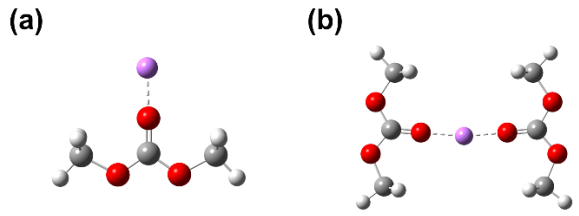


Figure S9. Optimized bonding structures of Li^+ and DMC. (a) Li^+–$DMC_{1(1)}$, (b) Li^+–$DMC_{2(1+1)}$. The subscript of each label indicates the number of DMC, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

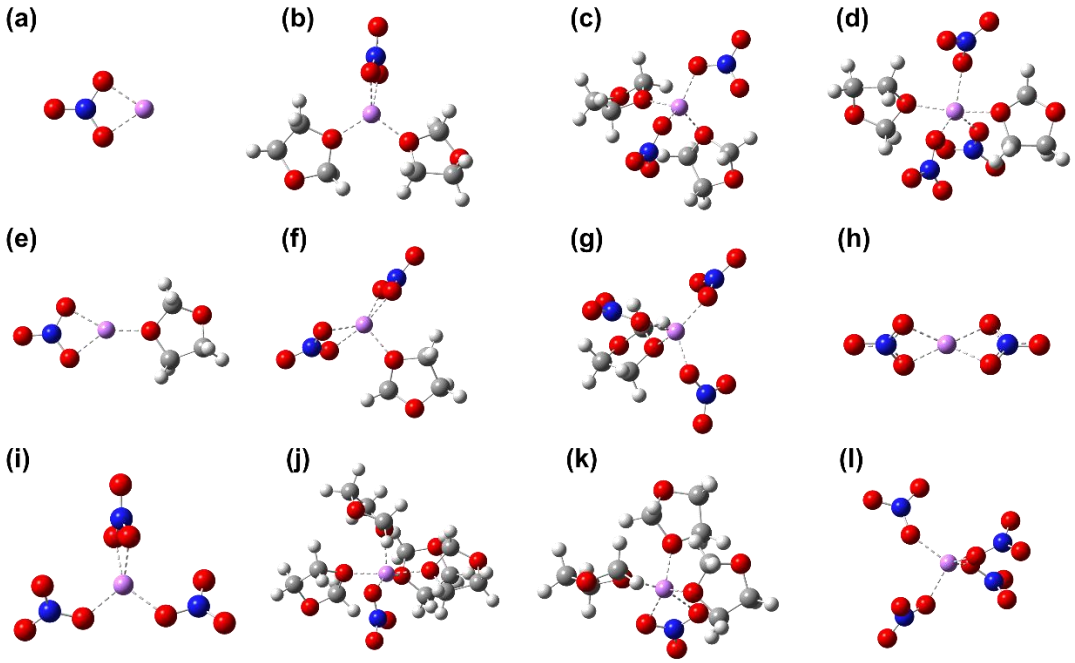


Figure S10. Optimized bonding structures of Li^+ and NO_3^- and/or DOL. (a) Li^+–$NO_3^-{}_{1(2)}$, (b) Li^+–$DOL_{2(1+1)}$–$NO_3^-{}_{1(2)}$, (c) Li^+–$DOL_{2(1+1)}$–$NO_3^-{}_{2(1+1)}$, (d) Li^+–$DOL_{2(1+1)}$–$NO_3^-{}_{3(1+1+1)}$, (e) Li^+–$DOL_{1(1)}$–$NO_3^-{}_{1(2)}$, (f) Li^+–$DOL_{1(1)}$–$NO_3^-{}_{2(2+2)}$, (g) Li^+–$DOL_{1(1)}$–$NO_3^-{}_{3(1+1+1)}$, (h) Li^+–$NO_3^-{}_{2(2+2)}$, (i) Li^+–$NO_3^-{}_{3(2+1+1)}$, (j) Li^+–$DOL_{4(1+1+1+1)}$–$NO_3^-{}_{1(1)}$, (k) Li^+–$DOL_{3(1+1+1)}$–$NO_3^-{}_{1(1)}$, (l) Li^+–$NO_3^-{}_{4(1+1+1+1)}$. The subscript of each label indicates the respective number of DOL and NO_3^-, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

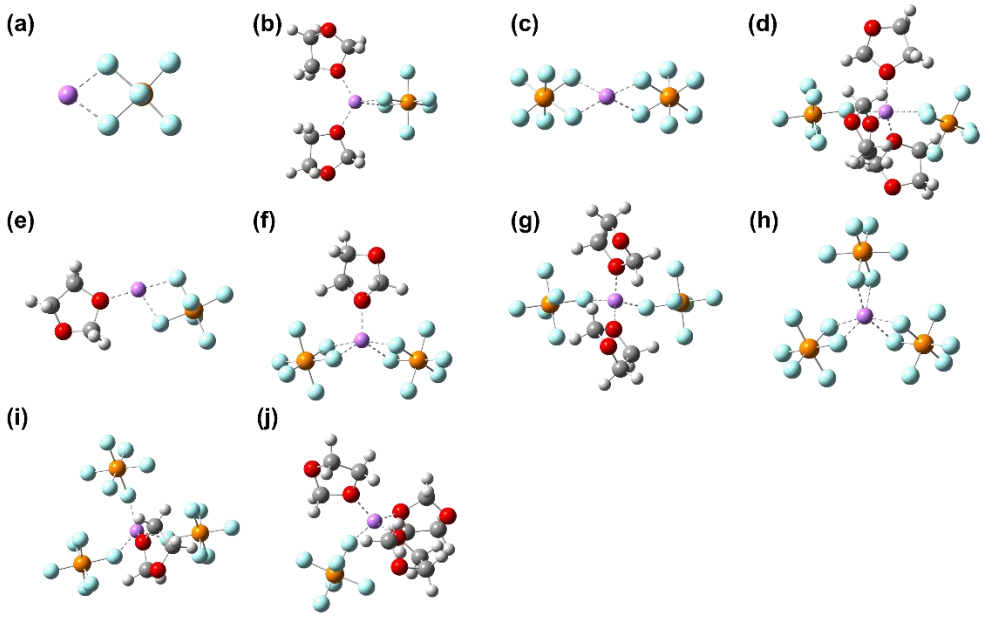


Figure S11. Optimized bonding structures of Li^+ and PF_6^- and/or DOL. (a) Li^+–$PF_6^-{}_{1(2)}$, (b) Li^+–$DOL_{2(1+1)}$–$PF_6^-{}_{1(2)}$, (c) Li^+–$PF_6^-{}_{2(2+2)}$, (d) Li^+–$DOL_{3(1+1+1)}$–$PF_6^-{}_{2(1+1)}$, (e) Li^+–$DOL_{1(1)}$–$PF_6^-{}_{1(2)}$, (f) Li^+–$DOL_{1(1)}$–$PF_6^-{}_{2(2+2)}$, (g) Li^+–$DOL_{2(1+1)}$–$PF_6^-{}_{2(1+1)}$, (h) Li^+–$PF_6^-{}_{3(2+2+2)}$, (i) Li^+–$DOL_{1(1)}$–$PF_6^-{}_{3(1+1+1)}$, (j) Li^+–$DOL_{3(1+1+1)}$–$PF_6^-{}_{1(1)}$. The subscript of each label indicates the respective number of DOL and PF_6^-, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

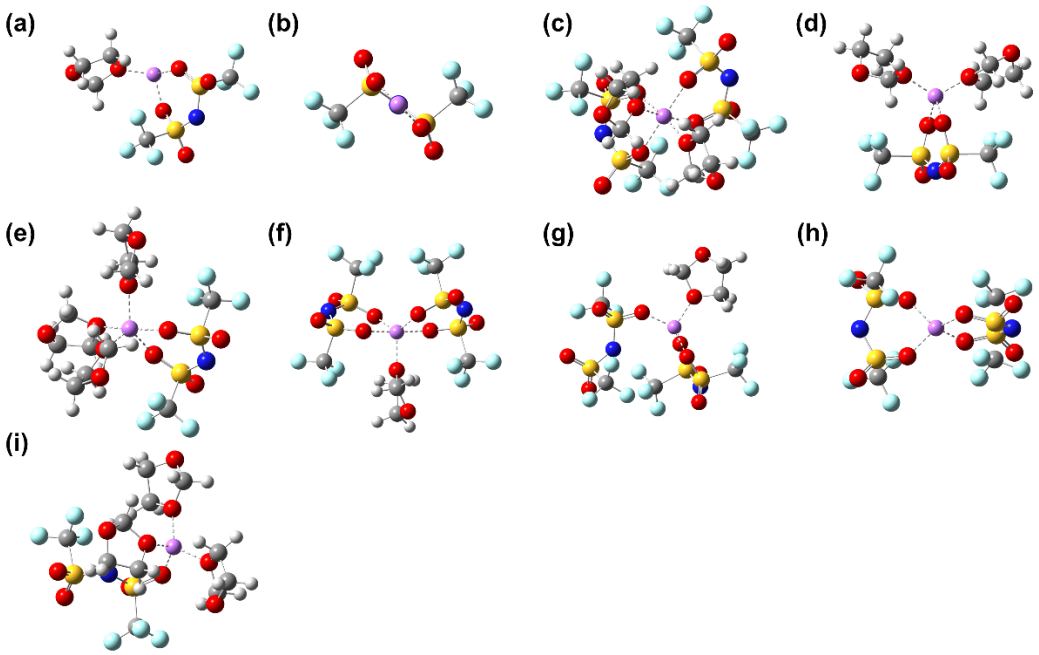


Figure S12. Optimized bonding structures of Li^+ and $TFSI^-$ and/or DOL. (a) Li^+–$DOL_{1(1)}$–$TFSI^-_{1(2)}$, (b) Li^+–$TFSI^-_{1(2)}$, (c) Li^+–$DOL_{2(1+1)}$–$TFSI^-_{2(2+2)}$, (d) Li^+–$DOL_{2(1+1)}$–$TFSI^-_{1(2)}$, (e) Li^+–$DOL_{3(1+1+1)}$–$TFSI^-_{1(2)}$, (f) Li^+–$DOL_{1(1)}$–$TFSI^-_{2(2+2)}$, (g) Li^+–$DOL_{1(1)}$–$TFSI^-_{2(2+1)}$, (h) Li^+–$TFSI^-_{2(2+2)}$, (i) Li^+–$DOL_{3(1+1+1)}$–$TFSI^-_{1(1)}$. The subscript of each label indicates the respective number of DOL and $TFSI^-$, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

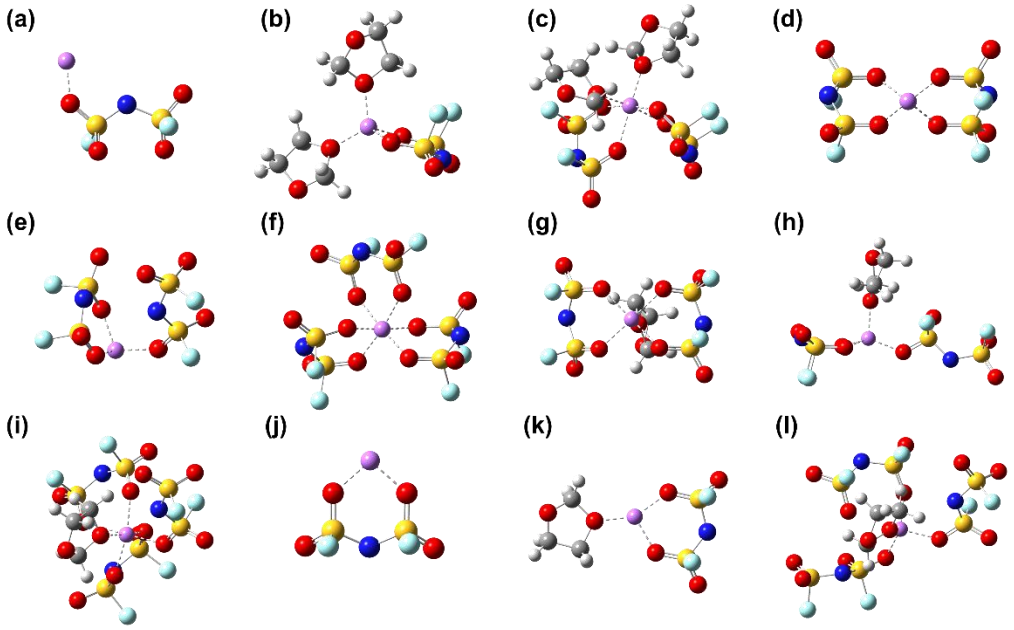


Figure S13. Optimized bonding structures of Li^+ and FSI^- and/or DOL. (a) Li^+–$FSI^-_{1(1)}$, (b) Li^+–$DOL_{2(1+1)}$–$FSI^-_{1(2)}$, (c) Li^+–$DOL_{2(1+1)}$–$FSI^-_{2(2+2)}$, (d) Li^+–$FSI^-_{2(2+2)}$, (e) Li^+–$FSI^-_{2(2+1)}$, (f) Li^+–$FSI^-_{3(2+2+2)}$, (g) Li^+–$DOL_{1(1)}$–$FSI^-_{2(2+2)}$, (h) Li^+–$DOL_{1(1)}$–$FSI^-_{2(2+1)}$, (i) Li^+–$DOL_{1(1)}$–$FSI^-_{3(2+2+1)}$, (j) Li^+–$FSI^-_{1(2)}$, (k) Li^+–$DOL_{1(1)}$–$FSI^-_{1(2)}$, (l) Li^+–$DOL_{1(1)}$–$FSI^-_{3(2+1+1)}$. The subscript of each label indicates the respective number of DOL and FSI^-, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

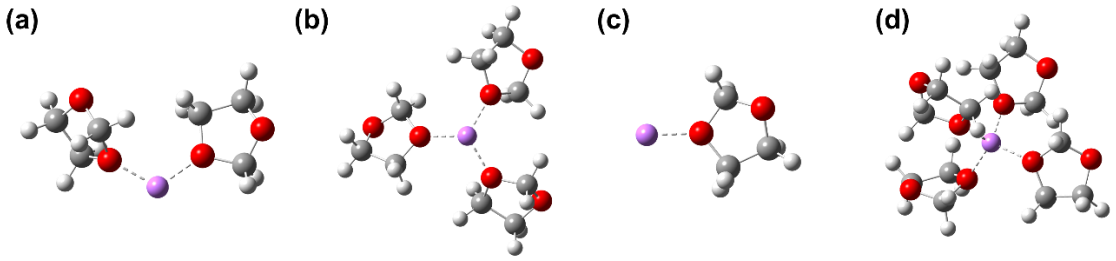


Figure S14. Optimized bonding structures of Li^+ and DOL. (a) Li^+–$DOL_{2(1+1)}$, (b) Li^+–$DOL_{3(1+1+1)}$, (c) Li^+–$DOL_{1(1)}$, (d) Li^+–$DOL_{4(1+1+1+1)}$. The subscript of each label indicates the number of DOL, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

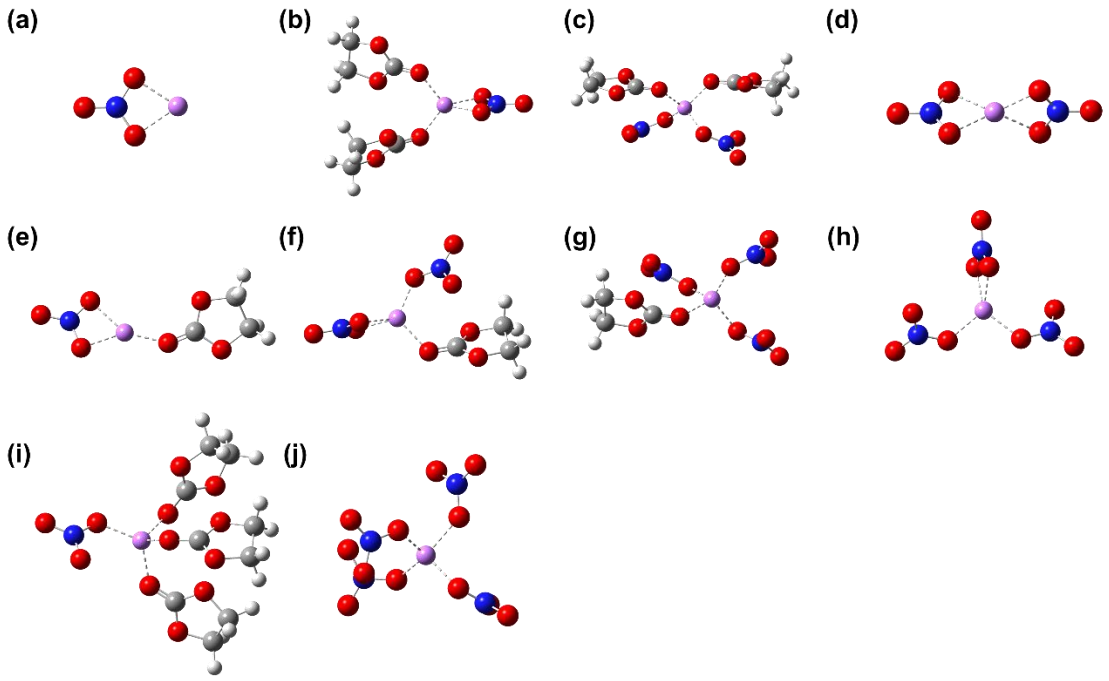


Figure S15. Optimized bonding structures of Li^+ and NO_3^- and/or EC. (a) Li^+–$NO_3^-{}_{1(2)}$, (b) Li^+–$EC_{2(1+1)}$–$NO_3^-{}_{1(2)}$, (c) Li^+–$EC_{2(1+1)}$–$NO_3^-{}_{2(1+1)}$, (d) Li^+–$NO_3^-{}_{2(2+2)}$, (e) Li^+–$EC_{1(1)}$–$NO_3^-{}_{1(2)}$, (f) Li^+–$EC_{1(1)}$–$NO_3^-{}_{2(2+1)}$, (g) Li^+–$EC_{1(1)}$–$NO_3^-{}_{3(1+1+1)}$, (h) Li^+–$NO_3^-{}_{3(2+1+1)}$, (i) Li^+–$EC_{3(1+1+1)}$–$NO_3^-{}_{1(1)}$, (j) Li^+–$NO_3^-{}_{4(1+1+1+1)}$. The subscript of each label indicates the respective number of EC and NO_3^-, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

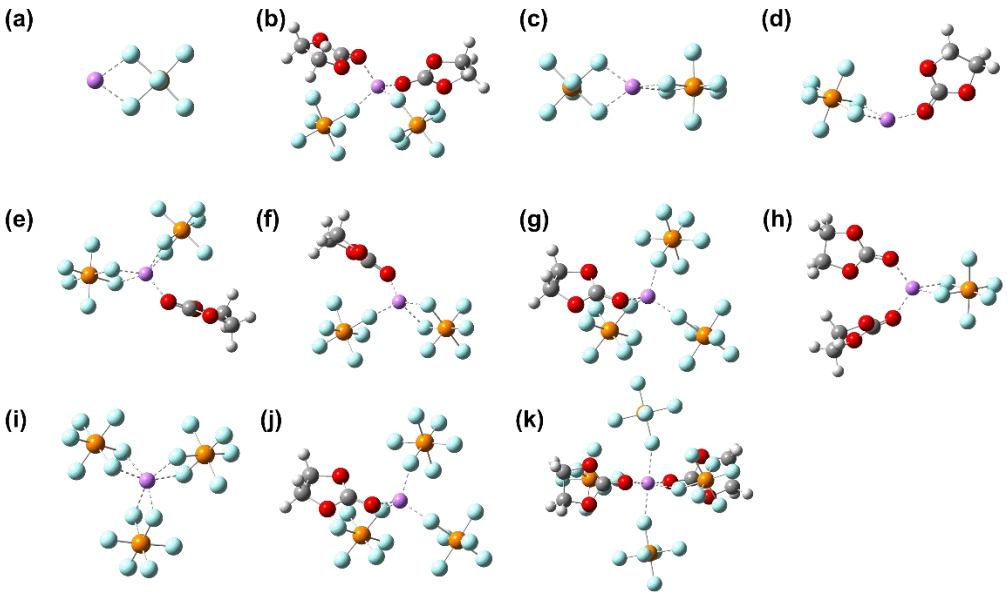


Figure S16. Optimized bonding structures of Li^+ and PF_6^- and/or EC. (a) Li^+–$PF_6^-{}_{1(2)}$, (b) Li^+–$EC_{2(1+1)}$–$PF_6^-{}_{2(1+1)}$, (c) Li^+–$PF_6^-{}_{2(2+2)}$, (d) Li^+–$EC_{1(1)}$–$PF_6^-{}_{1(2)}$, (e) Li^+–$EC_{1(1)}$–$PF_6^-{}_{2(2+2)}$, (f) Li^+–$EC_{1(1)}$–$PF_6^-{}_{2(2+1)}$, (g) Li^+–$EC_{1(1)}$–$PF_6^-{}_{3(1+1+1)}$, (h) Li^+–$EC_{2(1+1)}$–$PF_6^-{}_{1(2)}$, (i) Li^+–$PF_6^-{}_{3(2+2+2)}$, (j) Li^+–$EC_{1(1)}$–$PF_6^-{}_{3(1+1+1)}$, (k) Li^+–$EC_{2(1+1)}$–$PF_6^-{}_{4(1+1+1+1)}$. The subscript of each label indicates the respective number of EC and PF_6^-, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

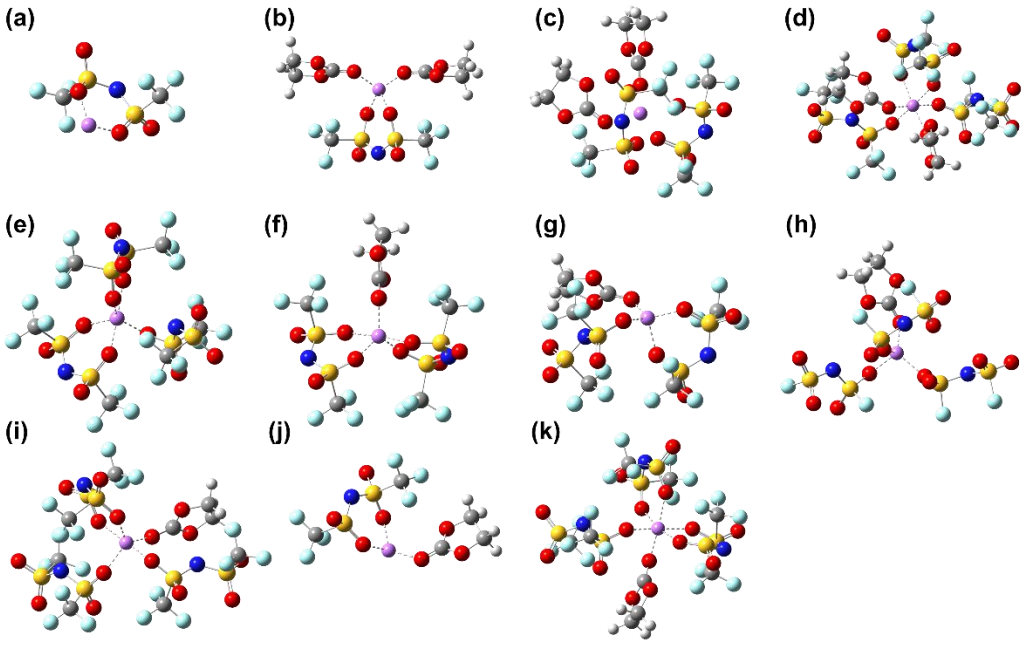


Figure S17. Optimized bonding structures of Li^+ and $TFSI^-$ and/or EC. (a) Li^+–$TFSI^-_{1(2)}$, (b) Li^+–$EC_{2(1+1)}$–$TFSI^-_{1(2)}$, (c) Li^+–$EC_{2(1+1)}$–$TFSI^-_{2(2+2)}$, (d) Li^+–$EC_{1(1)}$–$TFSI^-_{3(2+1+1)}$, (e) Li^+–$TFSI^-_{3(2+2+1)}$, (f) Li^+–$EC_{1(1)}$–$TFSI^-_{2(2+2)}$, (g) Li^+–$EC_{1(1)}$–$TFSI^-_{2(1+2)}$, (h) Li^+–$EC_{1(1)}$–$TFSI^-_{3(2+1+1)}$, (i) Li^+–$EC_{1(1)}$–$TFSI^-_{3(2+1+1)}$, (j) Li^+–$EC_{1(1)}$–$TFSI^-_{1(2)}$, (k) Li^+–$EC_{1(1)}$–$TFSI^-_{3(2+2+1)}$. The subscript of each label indicates the respective number of EC and $TFSI^-$, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

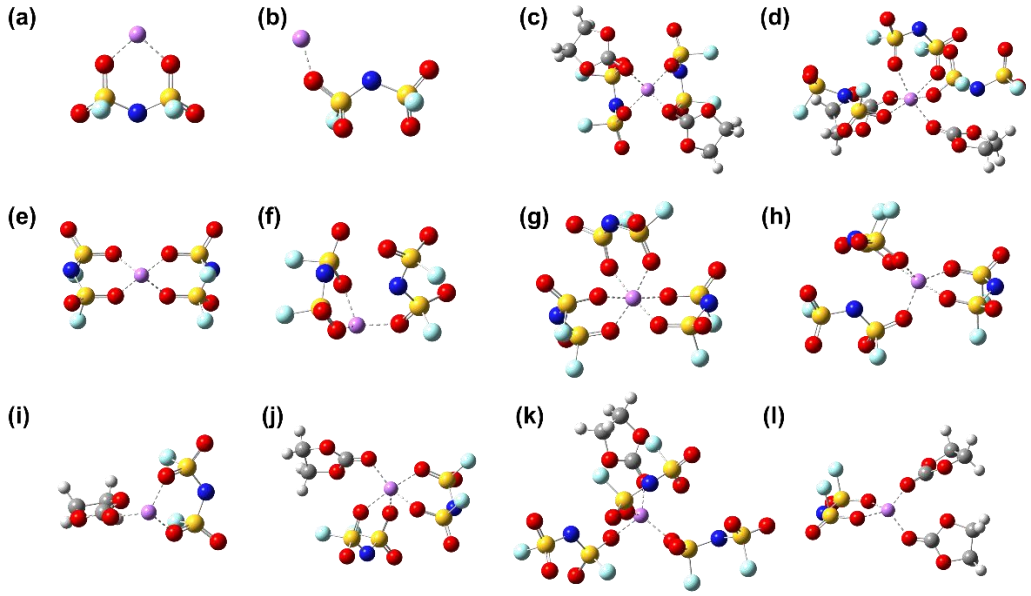


Figure S18. Optimized bonding structures of Li^+ and FSI and/or EC. (a) Li^+–$FSI^-_{1(2)}$, (b) Li^+–$FSI^-_{1(1)}$, (c) Li^+–$EC_{2(1+1)}$–$FSI^-_{2(2+2)}$, (d) Li^+–$EC_{2(1+1)}$–$FSI^-_{3(2+1+1)}$, (e) Li^+–$FSI^-_{2(2+2)}$, (f) Li^+–$FSI^-_{2(2+1)}$, (g) Li^+–$FSI^-_{3(2+2+2)}$, (h) Li^+–$FSI^-_{3(2+2+1)}$, (i) Li^+–$EC_{1(1)}$–$FSI^-_{1(2)}$, (j) Li^+–$EC_{1(1)}$–$FSI^-_{2(2+2)}$,, (k) Li^+–$EC_{1(1)}$–$FSI^-_{3(2+1+1)}$, (l) Li^+–$EC_{2(1+1)}$–$FSI^-_{1(2)}$. The subscript of each label indicates the respective number of EC and FSI^-, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

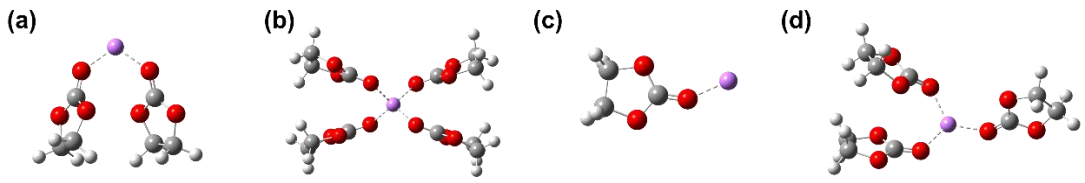


Figure S19. Optimized bonding structures of Li^+ and EC. (a) Li^+–$EC_{2(1+1)}$, (b) Li^+–$EC_{4(1+1+1+1)}$, (c) Li^+–$EC_{1(1)}$, (d) Li^+–$EC_{3(1+1+1)}$. The subscript of each label indicates the number of EC, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

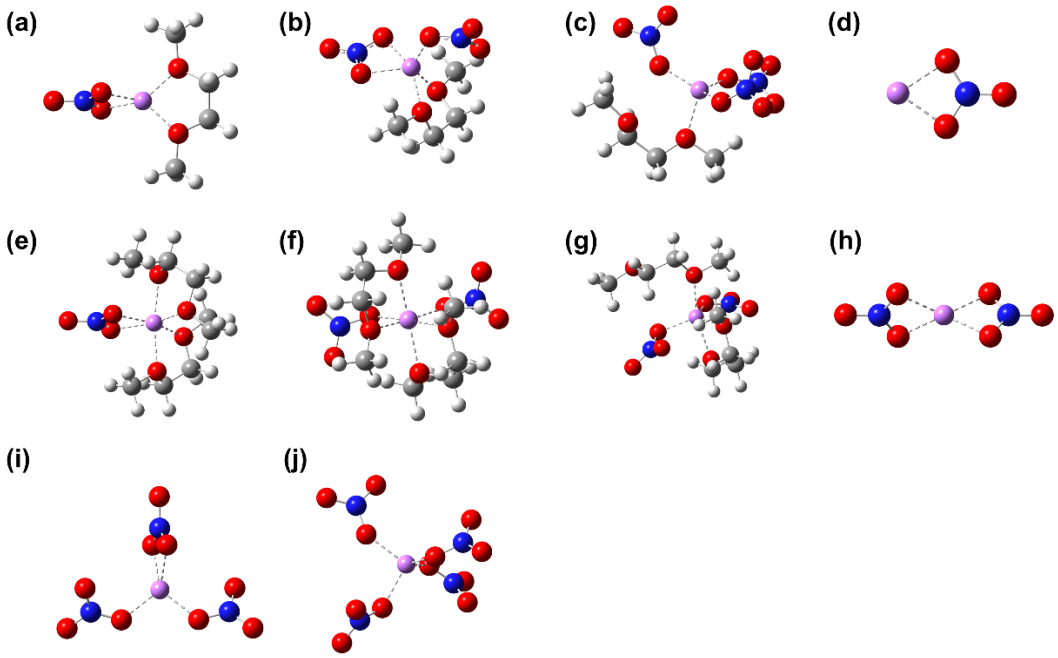


Figure S20. Optimized bonding structures of Li^+ and NO_3^- and/or DME. (a) Li^+–$DME_{1(2)}$–$NO_3^-{}_{1(2)}$, (b) Li^+–$DME_{1(2)}$–$NO_3^-{}_{2(2+1)}$, (c) Li^+–$DME_{1(1)}$–$NO_3^-{}_{3(1+1+1)}$, (d) Li^+–$NO_3^-{}_{1(2)}$, (e) Li^+–$DME_{2(2+2)}$–$NO_3^-{}_{1(2)}$, (f) Li^+–$DME_{2(2+2)}$–$NO_3^-{}_{2(1+1)}$, (g) Li^+–$DME_{2(1+1)}$–$NO_3^-{}_{2(1+1)}$, (h) Li^+–$NO_3^-{}_{2(2+2)}$, (i) Li^+–$NO_3^-{}_{3(2+1+1)}$, (j) Li^+–$NO_3^-{}_{4(1+1+1+1)}$. The subscript of each label indicates the respective number of DME and NO_3^-, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

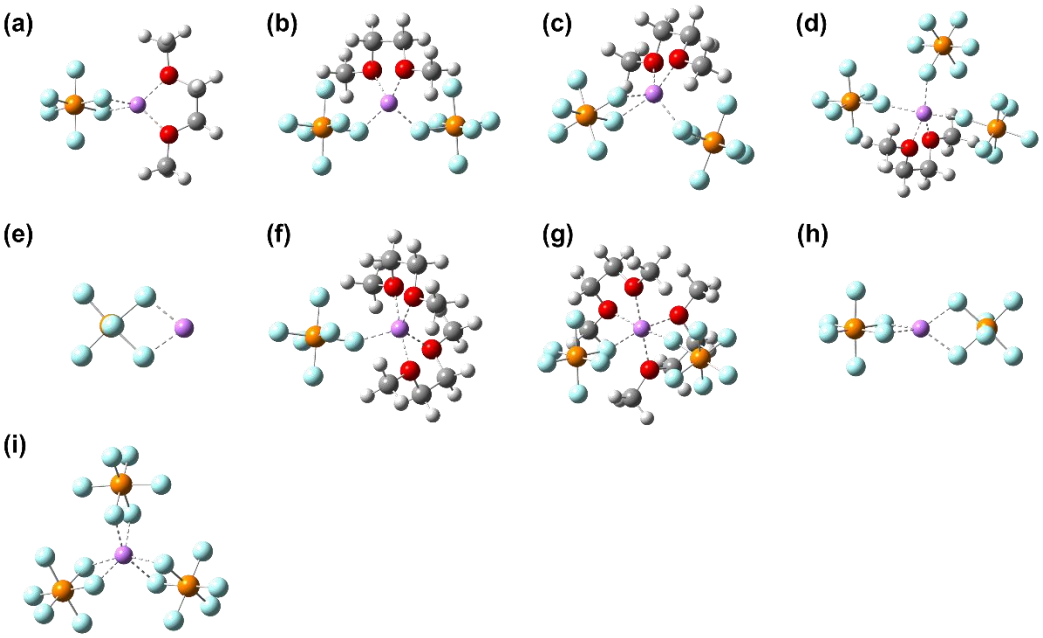


Figure S21. Optimized bonding structures of Li^+ and PF_6^- and/or DME. (a) Li^+–$DME_{1(2)}$–$PF_6^-{}_{1(2)}$, (b) Li^+–$DME_{1(2)}$–$PF_6^-{}_{2(1+1)}$, (c) Li^+–$DME_{1(2)}$–$PF_6^-{}_{2(2+1)}$, (d) Li^+–$DME_{1(1)}$–$PF_6^-{}_{3(1+1+1)}$, (e) Li^+–$PF_6^-{}_{1(2)}$, (f) Li^+–$DME_{1(1)}$–$PF_6^-{}_{2(2+2)}$, (g) Li^+–$DME_{2(2+2)}$–$PF_6^-{}_{2(1+1)}$, (h) Li^+–$PF_6^-{}_{2(2+2)}$, (i) Li^+–$PF_6^-{}_{3(2+2+2)}$. The subscript of each label indicates the respective number of DME and PF_6^-, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

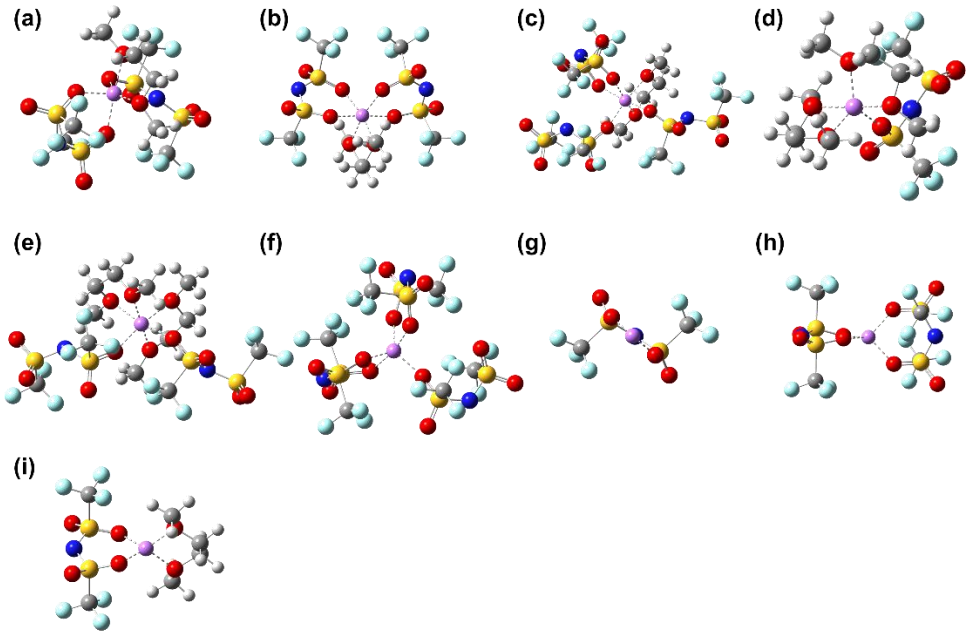


Figure S22. Optimized bonding structures of Li^+ and $TFSI^-$ and/or DME. (a) Li^+–$DME_{1(2)}$–$TFSI^-_{2(2+1)}$, (b) Li^+–$DME_{1(2)}$–$TFSI^-_{2(2+2)}$, (c) Li^+–$DME_{1(2)}$–$TFSI^-_{3(2+1+1)}$, (d) Li^+–$DME_{2(2+2)}$–$TFSI^-_{1(1)}$, (e) Li^+–$DME_{2(2+2)}$–$TFSI^-_{2(1+1)}$, (f) Li^+–$TFSI^-_{3(2+2+1)}$, (g) Li^+–$TFSI^-_{1(2)}$, (h) Li^+–$TFSI^-_{2(2+2)}$, (i) Li^+–$DME_{1(2)}$–$TFSI^-_{1(2)}$. The subscript of each label indicates the respective number of DME and $TFSI^-$, and the numbers in brackets represent the mono- (1) or bi-dentate (2) of each bonding molecule. Atomic color: H-while, Li-violet, C-gray, N-blue, O-red, F-cyan, P-orange, S-yellow.

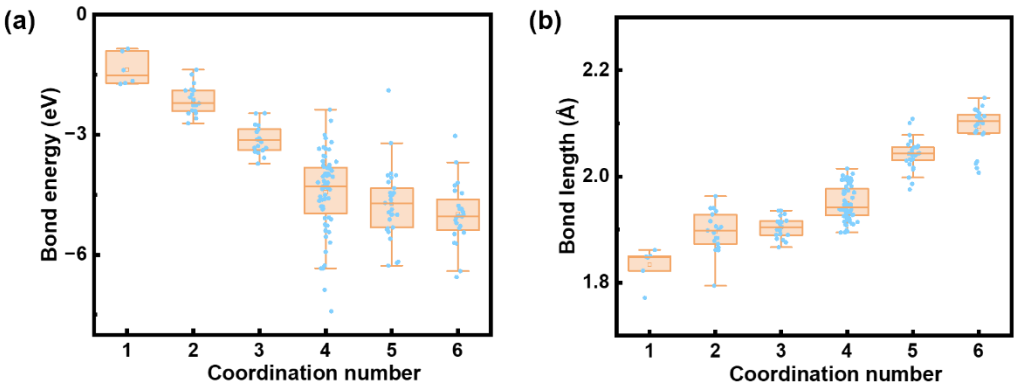


Figure S23. Changes of bond energy (a) and bond length (b) with different coordination numbers.

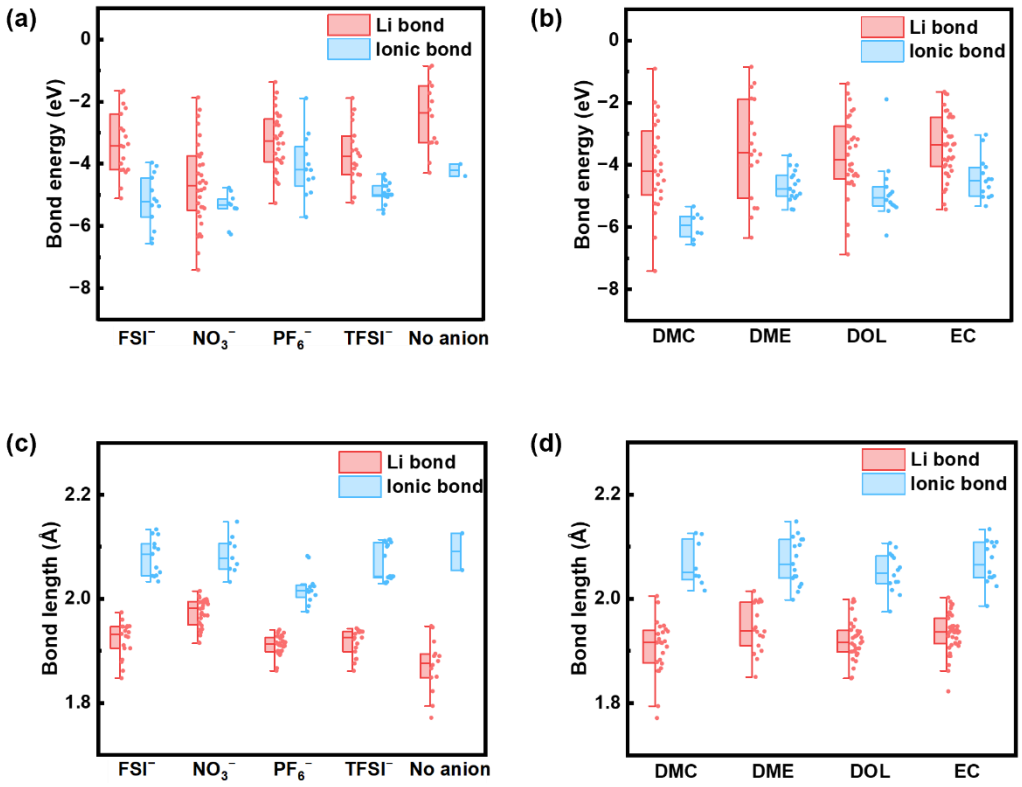


Figure S24. Li bond and ionic bond analysis based on Li–O/F bond length and bond energy. (a) Distribution of Li–O/F bond energy for different anions ($TFSI^-$, FSI^-, PF_6^-, NO_3^-, and no anion coordinated), highlighting differences between Li bonds and ionic bonds. (b) Distribution of Li–O/F bond energy for different solvents (DMC, DME, DOL, and EC). (c) Distribution of Li–O/F bond length for different anions ($TFSI^-$, FSI^-, PF_6^-, NO_3^-, and no anion coordinated). (d) Distribution of Li–O/F bond length for different solvents (DMC, DME, DOL, and EC). Color: Li bond—red, ionic bond—blue.

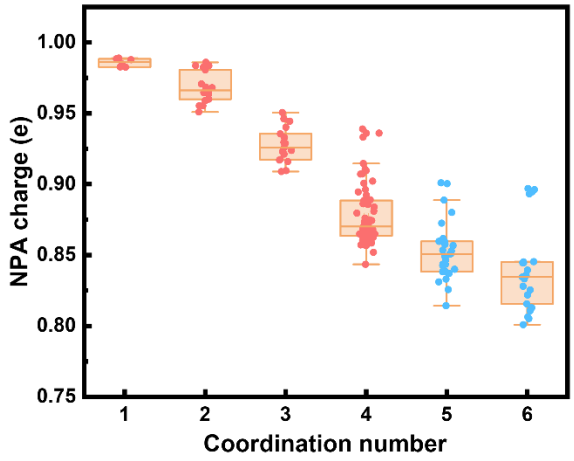


Figure S25. The variation of the atomic charge based on the natural population analysis (NPA charge) with the change of the coordination number. Color: Li bond—red, ionic bond—blue.

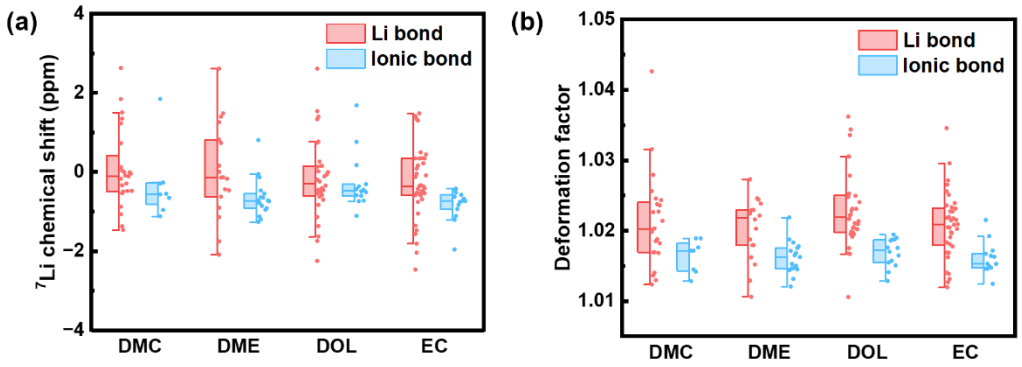


Figure S26. Distribution of ^{7}Li chemical shifts (a) and ELF deformation factors (b) for different solvents (DMC, DME, DOL, and EC). Color: Li bond—red, ionic bond—blue.

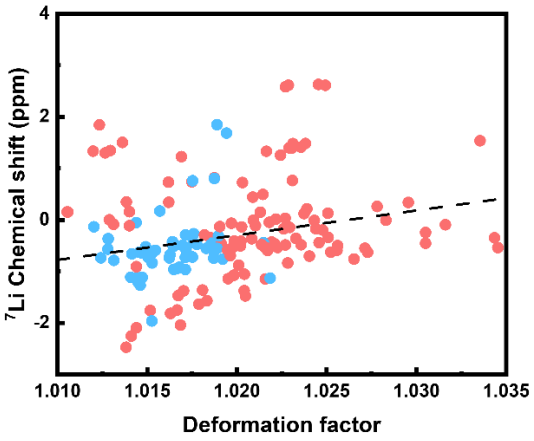


Figure S27. The variation of the ^{7}Li chemical shifts with the change of ELF deformation factors, highlighting differences between Li bonds and ionic bonds. Color: Li bond—red, ionic bond—blue.

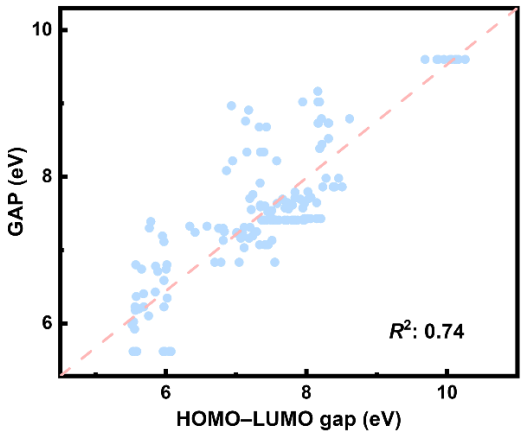


Figure S28. The variation of the machine learning feature GAP with the change of the HOMO–LUMO gap of the cluster.



Figure S29. The Pearson correlation coefficient analysis for machine learning features.

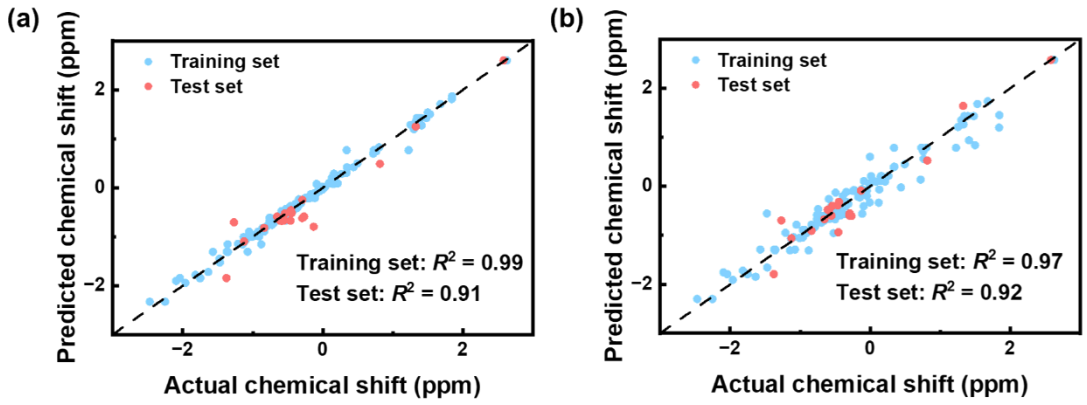


Figure S30. Comparison of single machine learning models against the ensemble model on performance. (a) Prediction of ^{7}Li chemical shift made by GBR model versus actual chemical shift on training and test set. (b) Prediction of ^{7}Li chemical shift made by SVR model versus actual chemical shift on training and test set.

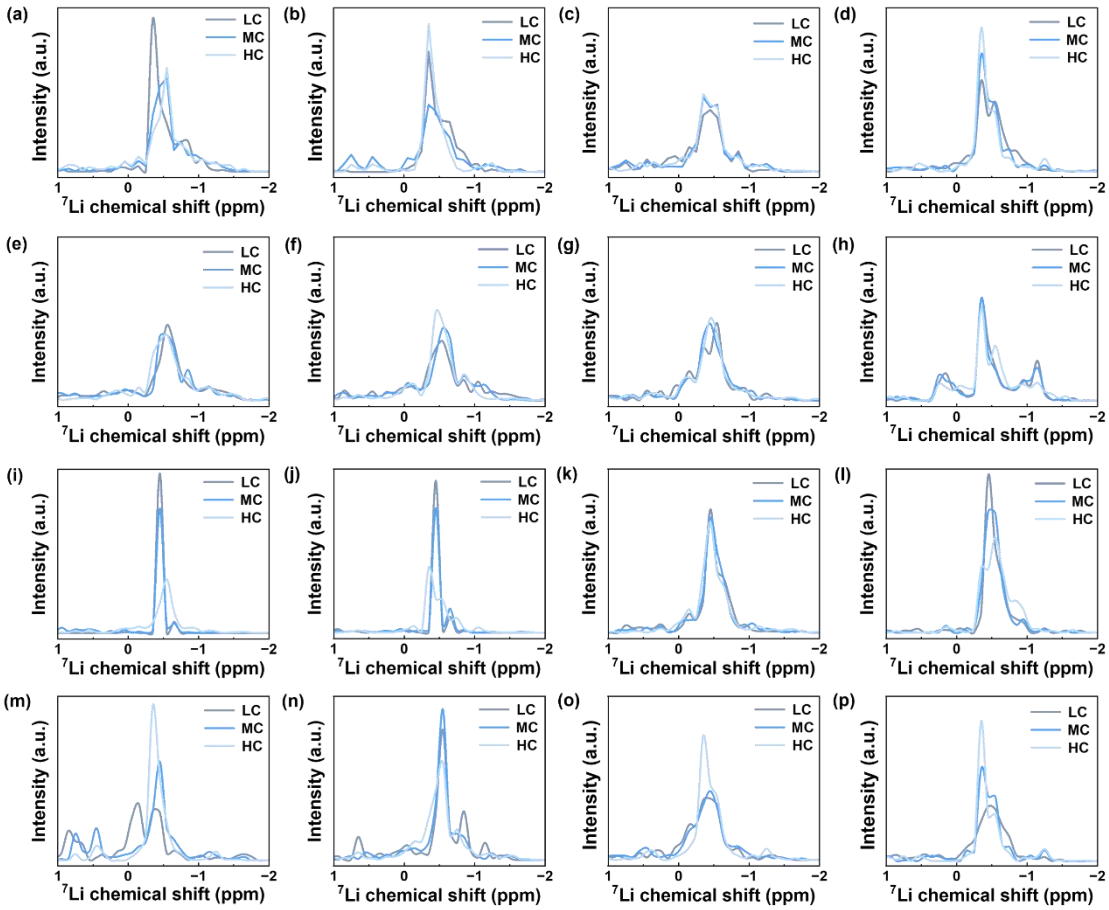


Figure S31. Distribution of predicted ^{7}Li chemical shift for LC, MC, and HC electrolytes. (a–d) Distribution of predicted ^{7}Li chemical shift for LC, MC, and HC (a) EC/TFSI^{-}, (b) EC/PF_6^{-}, (c) EC/NO_3^{-}, and (d) EC/FSI^{-} electrolytes. (e–h) Distribution of predicted ^{7}Li chemical shift for LC, MC, and HC (e) DOL/TFSI^{-}, (f) DOL/PF_6^{-}, (g) DOL/NO_3^{-}, and (h) DOL/FSI^{-} electrolytes. (i–l) Distribution of predicted ^{7}Li chemical shift for LC, MC, and HC (i) DME/TFSI^{-}, (j) DME/PF_6^{-}, (k) DME/ NO_3^{-}, and (l) DME/FSI^{-} electrolytes. (m–p) Distribution of predicted ^{7}Li chemical shift for LC, MC, and HC (m) DMC/TFSI^{-}, (n) DMC/PF_6^{-}, (o) DMC/NO_3^{-}, and (p) DMC/FSI^{-} electrolytes.

III. Supplemental Tables

Table S1. Comparisons of the Li bond and the ionic bond

Characteristic	Li bond	Ionic bond
CN	1–4	5–6
Average bond length	~1.9 Å (shorter)	~2.1 Å (longer)
Average bond energy	~−3.7 eV (weaker)	~−4.9 eV (stronger)
^{7}Li NMR shift trend (with increasing CN)	Downfield (deshielding)	Upfield (shielding)
ELF deformation factor trend (with increasing CN)	Increases (more asymmetric)	Decreases (more symmetric)
Average lifespan in electrolyte (with increasing salt concentration)	Increases	No clear trend

Table S2. Explanations of machine learning features

Feature	Explanation
CNA	Coordination number of anions
CNS	Coordination number of solvent molecules
NCA	Number of coordinating anions in the cluster
NCS	Number of coordinating solvent molecules in the cluster
GAP	weighted averaging the HOMO–LUMO gaps of coordinating solvent molecules or anions, with weights corresponding to their numbers in the cluster $\mathrm{GAP} = \frac{\mathrm{NCA}}{\mathrm{NCA}+\mathrm{NCS}}(\mathrm{HOMO}-\mathrm{LUMO})_{\mathrm{anion}} + \frac{\mathrm{NCS}}{\mathrm{NCA}+\mathrm{NCS}}(\mathrm{HOMO}-\mathrm{LUMO})_{\mathrm{solvent}}$

IV. Supporting References

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